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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
UNIT 4- Multithreaded Programming								
114	4	Select the method that should be overridden to provide thread's behavior when Thread class is extended:	D	1	start()	resume()	runnable()	run()
115	4	What are Threads?	A	1	lightweight process	heavyweight process	Both	none of above
116	4	Multithreading is also called as _____	C	1	Recurrent	Crosscurrent	Concurrency	Simultaneity
117	4	What is the default relation b/w Thread & Runnable?	D	1	Thread is child of Runnable	Runnable extends Thread	Thread extends Runnable	Thread implements Runnable
118	4	Which method is used to get the object of current thread?	B	1	Thread.CurrentThread()	Thread.currentThread()	Thread.liveThread()	Thread.getThread()
119	4	Which method throws InterruptedException	A	1	join()	start()	run()	currentThread()
120	4	Two threads cannot simultaneously enter into the methods of the same object if the methods are	B	1	static	synchronized	synchronize	synchronization
121	4	Which mathod is not a part of multi-threading?	A	1	Sleep()	join()	wait()	run()
122	4	mathods used for inter thread communication are?	D	1	wait()	notify()	notifyAll()	All of the above
123	4	Which of these are types of multitasking?	C	1	Process based	Thread based	Process and Thread based	None of the mentioned
124	4	If there occurs any exception in thread, then other threads	C	1	stop executing	gets impacted	doesn't gets impacted	daemon thread starts executing
125	4	Which mathod is used to assign name of thread by programmer?	D	1	run()	start()	getName()	setName(String s)
126	4	Which of these method waits for the thread to terminate?	A	1	join()	stop()	sleep()	isAlive()
127	4	Which of the following method is not available in Thread class?	D	1	join()	run()	sleep()	Run()
128	4	Which function of pre defined class Thread is used to check weather current thread being checked is still running?	C	1	Join()	Alive()	isAlive()	isRunning()
129	4	Which of these method is used to tell the calling thread to give up a monitor and go to sleep until some other thread enters the same monitor?	A	1	wait()	sleep()	notifyAll()	notify()
130	4	What is true about threading?	D	1	run() method creates new thread	run() method calls start() method and runs the code	run() method can be called directly without start() method being called	start() method creates new thread and calls code written in run() method
131	4	Thread synchronization in a process will be required when	D	1	All threads sharing the same address space	All threads sharing the same global variables	All threads sharing the same files	All
132	4	Which of the following is a correct way to synchronize a method in Java code?	A	1	public synchronized void method() { }	public void method() { synchronized { } }	public void synchronized method() { }	public void method() { synchronized void() { } }
133	4	Answer the Question : start() method is called to _____ ?	A	1	Invoke the thread, before the run() method is executed	Execute the thread which is in 'ready to run' state	Transit the thread from 'running' to 'ready to run' state	Transit from runnable to waiting state
134	4	Threads in java share the same memory space and resources.	B	1	FALSE	TRUE	Depends on Compiler	Depends on JVM

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135	4	We can create thread in java by a) Implementing Thread Class b) Implementing Runnable Interface c) Implementing Thread Interface d) Extending Thread class e) Extending Runnable class	B	1	a & b only	b & d only	a & e only	c & e only
136	4	The tasks or job that thread needs to perform is written inside	D	1	inner class	Static Block	Constructor	run()
137	4	Which is the correct Thread Class constructor? A) Thread(Runnable r) B) Thread(Runnable r, Thread t) C) Thread(Runnable r, String s) D) Thread(String s, Runnable r)	C	1	A & B only	C & D only	A & C Only	A & D Only

138	4	What is the name of the thread in output of this program?
		Class A
		{
		public static void main(String args[])
		{
		Thread t = Thread.currentThread();
		System.out.println(t.getName());
		}
		}
			A	1	main	java	system	thread

139	4	what is output of this code?
		class A extends Thread {
		public void run() {
		System.out.println("Hello");
		}
		public void start() {
		System.out.println("Hii");
		}
		}
		class LJU {
		public static void main(String[] args) {
		A a = new A();
		a.start();
		}
		}
			B	1	Hello	Hii	HelloHii	None of the above

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

140	4	what is the output of this code?
		class A extends Thread {
		public void run() {
		System.out.println("Hello");
		}
		public void run(int a) {
		System.out.println("Hii");
		}
		}
		class LJU {
		public static void main(String[] args) {
		A a = new A();
		a.start(10);
		}
		}
			D	1	Hello	Hii	HelloHii	Error

141	4	class multithreaded_programing {
		public static void main(String args[]) {
		Thread t = Thread.currentThread();
		Thread t1 = t;
		System.out.println(t1);
		}
		}
			C	1	Thread[0,main]	Thread[t,5,main]	Thread[main,5,main]	Thread[t1,5,main]

142	4	Assume the following method is properly synchronized and called from a thread A on an object B: wait(2000); After calling this method, when will the thread A become a candidate to get another turn at the CPU?	C	1	Two seconds after thread A is notified	Two seconds after lock B is released	After thread A is notified or after two seconds.	After the lock on B is released and after two seconds.
143	4	Which of these statement is incorrect?	D	1	start() method is used to begin execution of the thread	A thread can be formed by a class that extends thread class	A thread can be formed by implementing Runnable interface only	run() method is used to begin execution of a thread before start() method in special cases

144	4	What will be the output of the following code?
		class LJ extends Thread {
		public static void main(String[] args) {
		LJ t = new LJ();
		Thread x = new Thread(t);
		x.start(); /* Line 7 */
		}
		public void run() {
		for (int i = 0; i < 3; ++i) {
		System.out.print(i + ".."); } }
		}
			A	1	0..1..2..	0..1..2..3..	1..2..3..	Compilation fails

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

145	4	Which of the following line of code is suitable to start a thread ?
		class X implements Runnable
		{
		public static void main(String args[])
		{
		/* Missing code? */
		}
		public void run() {}
		}
			C	1	Thread t = new Thread(X);	Thread t = new Thread(X); t.start();	X run = new X();
							Thread t = new Thread(run); t.start();
								Thread t = new Thread();
								x.run();

146	4	What will be the output of the follwing code?
		class MyThread extends Thread
		{
		public static void main(String [] args)
		{
		MyThread t = new MyThread();
		t.start();
		System.out.print("one. ");
		t.start();
		System.out.print("two. ");
		}
		public void run() {
		System.out.print("Thread ");
		}
		}
			B	1	Compile time error	Run time exception	Thread one. Thread two.	one. two. Thread

147	4	What is output of this code?
		class A extends Thread {
		public void run() {
		System.out.println("Hello");
		}
		public void run(int a) {
		System.out.println("Hii");
		}
		}
		class LJU {
		public static void main(String[] args) {
		A a = new A();
		a.run(10);
		}
		}
			B	1	Hello	Hii	HelloHii	None of the above

148	4	What is the difference between synchronized methods and synchronized blocks in Java?	C	1	Synchronized methods can be used on non-static methods, while synchronized blocks can only be used on static methods	Synchronized methods are less efficient than synchronized blocks	Synchronized blocks can be used to synchronize access to specific parts of a method, while synchronized methods lock the entire method	There is no difference between synchronized methods and synchronized blocks

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149	4	What is the output?
		class TestDemo{
		public static void main(String args[])
		{
		Thread t = new Thread(“ONE”);
		Thread t2 = new Thread(“TWO”);
		Thread t3 = Thread.currentThread();
		System.out.println(t3.getName());
		}
		}
			C	1	ONE	TWO	main	No Output

150	4	class newthread extends Thread {
		Thread t;
		String name;
		newthread(String threadname) {
		name = threadname;
		t = new Thread(this, name);
		t.start(); }
		public void run() {}
		}
		class LJ {
		public static void main(String args[]) {
		newthread obj1 = new newthread("one");
		newthread obj2 = new newthread("two");
		try {
		obj1.t.wait();
		System.out.print(obj1.t.isAlive());
		} catch (Exception e) {
		System.out.print("Main thread interrupted");
		}
		}
		}
			C	1	TRUE	FALSE	Main thread interrupted	None of the mentioned

151	4	what is output of Below Java Programs?
		public class ThreadExtended extends Thread {
		public void run() {
		System.out.println("Thread is running no");
		}
		public static void main(String[] args)
		{
		ThreadExtended threadE = new ThreadExtended();
		ThreadExtended threadEx = new ThreadExtended();
		ThreadExtended threadEx1 = new ThreadExtended();
		threadEx.start();
		}
		}
			A	1	Thread is running no	Thread is running no
						Thread is running no
							Thread is running no
							Thread is running no
							Thread is running no
								None of the above

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

152	4	what is output of Below Java Programs?
		public class ThreadExtended extends Thread {
		public void run() {
		System.out.println(ThreadExtended.currentThread().getName());
		}
		public static void main(String[] args)
		{
		ThreadExtended threadE = new ThreadExtended();
		ThreadExtended threadEx = new ThreadExtended();
		ThreadExtended threadEx1 = new ThreadExtended();
		threadEx.start();
		}
		}
			D	1	4	2	Thread-0	Thread-1

153	4	The static method Thread.currentThread() returns a reference to the currently executing Thread object. What is the result of this code?
		class Test {
		public static void main(String [] args) {
		printAll(args);
		}
		public static void printAll(String[] lines) {
		for(int i = 0; i < lines.length; i++)
		{
		System.out.println(lines[i]);
		Thread.currentThread().sleep(1000);
		}
		}
		}
			D	1	Each String in the array lines will output, and there is no guarantee there will be a pause because currentThread() may not retrieve this thread.	Each String in the array lines will output, with no pause in between because this method is not executed in a Thread.	Each String in the array lines will output, with a 1-second pause.	This code will not compile.

154	4	What is the output of the following code?
		
		class LJU extends Thread {
		
		public void run(){
		System.out.print("Hi ");
		}
		}
		class Test{
		public static void main(String args[]) {
		LJU lj = new LJU();
		lj.run();
		System.out.println("Hello");
		}
		}
			B	1	HelloHi	Hi Hello	compile time error	output will be different every time

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155	4	What will be the output of the following code?
		class One extends Thread {
		public void run(){
		for(int i=0; i<2; i++)
		{
		System.out.print(i+" ");
		}
		}
		}
		class Test{
		public static void main(String args[])
		{
		Test t = new Test();
		One o = new One();
		t.call(o);
		}
		public void call(One o) {
		o.start();
		} }
			B	1	Compile time error	0 1	0 0	null null

156	4	What will be the output of the following code?
		class LJ extends Thread {
		public void run() {
		try {
		Thread.sleep('a');
		} catch (Exception e) {
		e.printStackTrace();
		}
		System.out.print("LJ"); } }
		class runLJ {
		synchronized public static void main(String[] args) throws Exception {
		LJ t1 = new LJ( );
		t1.start();
		synchronized(t1) {
		t1.wait();
		}
		System.out.println("U"); } }
			A	1	LJU	“LJU”	LJ	No Output

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157	4	What will be the output of the following code?
		class LJIET extends Thread {
		public void runn() {
		System.out.print('A' + 5);
		display();
		}
		static void display() {
		System.out.print("Q");
		} }
		class runLJ {
		public static void main(String[] args) throws Exception {
		LJIET t1 = new LJIET( );
		t1.start(); t1.join();
		System.out.println('a' + 3);
		} }
			D	1	Q100	70Q100	a3	100

158	4	What will be the output of the following code?
		class thread implements Runnable {
		static Thread m;
		public void run(){
		try {
		m.join( ); }
		catch (Exception e) {
		System.out.print("X"); }
		System.out.print("AB"); } }
		class runLJ {
		public static void main(String[] args) throws Exception {
		Thread t1 = new Thread(new thread ( ) , "Y");
		thread.m = Thread.currentThread();
		t1.start(); t1.join(1000);
		System.out.print("m"); } }
			A	1	mAB	ABm	XAB	Compile time error

159	4	What is the output of the Java code snippet?
		class Thread implements Runnable {
		public static void main(String[] args) {
		Thread t = new Thread();
		t.start();
		}
		public void run() {
		for (int i = 0; i < 2; ++i) {
		System.out.print("pop "); }
		}
		public void run(int number) {
		for (int i = 0; i < number; ++i) {
		System.out.print("push "); }
		} }
			C	1	pop pop	pop pop push push	Compile Error	Runtime Exception

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160	4	What is the output of the Java code snippet?
		
		class Test extends Thread {
		public static void main(String[] args) throws InterruptedException {
		new Thread (new Test( ),"true").start( );
		}
		synchronized public void run( ){
		Thread.currentThread( ).setName("TRUE");
		System.out.print(this.getName( ));
		}
		}
			B	1	main	Thread-0	Compile Error	Runtime Exception

161	4	What is the output of the Java code snippet?
		class Wait extends Thread implements Runnable {
		public static void main(String[] args) {
		Wait run = new Wait();
		Thread stop =new Thread(run);
		run.start();
		}
		public void run( ){
		System.out.print("Join ");
		}
		public void start( ){
		System.out.print("Sleep ");
		}
		}
			A	1	Sleep	Join Sleep	Compile Error	Runtime Exception

162	4	Write an application that creates and starts three threads. Each thread is instantiated from the same class. It executes a loop with 10 iterations. Each iteration displays string "HELLO", sleeps for 300 milliseconds. The application waits for all the threads to complete & displays the message "Good Bye..."		3				
163	4	Write a program to create a child thread to print integer numbers 1 to 10		3				
164	4	Write a program to create two threads, one thread will check whether given number is prime or not and second thread will print prime numbers between 0 to 100		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
165	4	It's IPL 2023 time. In this process of IPL-23 there is a thread called Bowler. This Bowler is going to create two threads with name – “Yorker” and “googly”. Write a java program to perform following task with each thread. Task 1 : if the thread name is "yorker" then it should print "SIXER" Task 2 : if the thread name is "googly" then it should print "WICKET" Make sure first thread bowl must be yorker. When execution of first thread is completed then and only then second bowl googly should work. Hint: Here Bowler is a Thread Class. and It must have execution of SIXER AND WICKET. Both yorker and googly must be from Bowler Thread Only. Give name IPL to the class of Main method.		5				
166	4	Write an application that executes two threads. One thread displays "Good Morning" every 1000 milliseconds & another thread displays "Good Afternoon" every 3000 milliseconds for 10 times.. Create the threads by implementing the Runnable interface.		5				
167	4	Write a program to create two threads, one thread will print odd numbers and second thread will print even numbers between 1 to 100 numbers		4				
168	4	Write a complete multi-threaded program to meet following requirements: o Two threads of same type are to be instantiated in the method main. o Each thread acts as a producer as well as a consumer. o A shared buffer can store only one integer information along with the source & destination of the information at a time. o The information produced is to be consumed by appropriate consumer. o Both producers produce information for both consumers. o Each thread produces 5 information.		5				
169	4	Create PaymentThread class by extending Thread and Make a program to count payments of 10 tickets each of 100 rs. Display total using main thread.		5				
170	4	Write a multithreaded program to print all odd positive numbers in ascending order up to n, where n is a positive integer number given as a command line argument. Instantiate requited number of threads, where each thread except the last, examines next 50 numbers and the last thread examines remaining numbers up to n		5				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
171	4	Write an application that read limit from user and executes two threads. One thread displays total of first n even numbers & another thread displays total of first n odd numbers. Create the threads by implementing the Runnable interface		4				
172	4	Write a program to create two thread one display alphabet from a to z and other will display numbers from 1 to 100		3				
173	4	write a program that demonstrates the prodecer consumer concept in detail.		4				
174	4	Explain the concept of Syncronization by taking suitable example.		4				
175	4	Write a complete multi-threaded program to meet following requirements: - Read matrix [A] m x n - Create m number of threads - Each thread computes summation of elements of one row, i.e. ith row of the matrix is processed by ith thread. Where 0 <= i< m. - Print the results		4				
176	4	Create a program for class PVR having Total_seats as instance variable.Create method for seat booking.Create two threads which try to book seat of PVR screen1 (common object). Use concept of synchronization.		6				
177	4	Make caller thread by extending thread class which calls synchronized receiver method available in receiver class. Make two threads of Caller and display message "Ringing" and thread name according to thread. Thread name must be Caller1 and Caller2.		6				
178	4	Implement a Prime Number Finder Write a Java program to implement a prime number finder that checks if a given number is prime or not in a separate thread. The program should use the join() method to wait for all the threads to finish, and the isAlive() method to check if a thread is still running. To solve this problem, you can create a prime number finder thread that implements the Runnable interface and checks if a given number is prime or not. You can then create multiple instances of the prime number finder thread and start them using separate threads. The main thread can use the join() method to wait for all the threads to finish, and the isAlive() method to check if a thread is still running.		5				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
179	4	Problem Statement: Implement a Bank Account Write a Java program to implement a bank account that allows multiple threads to deposit and withdraw money from the same account. The problem requires synchronization between the threads to prevent race conditions and ensure that the balance is accurate. To solve this problem, you can create a BankAccount class that has a balance variable and synchronized methods for depositing and withdrawing money. You can then create multiple threads that use the deposit and withdraw methods to modify the account balance.		5				
UNIT 5- Collection-Part 1								
180	5	Which of these class object can be used to form a dynamic array?	B	1	Array	ArrayList & Vector Both	Only Vector	Only ArrayList
181	5	What does get(int index) method define by List interface do ?	C	1	stores an object at the specified index	delete the object at specified index	returns the object at specified index	returns the object at specified index and modifies it's value

182	5	What will be the output of the following code?
		
		import java.util.ArrayList;
		class IPL{
		public static void main(String args[])
		{
		ArrayList<String> gt1 = new ArrayList<String>();
		gt1.addList("Hardik");
		gt1.addList("Saha");
		gt1.addList("Rahul");
		gt1.remove(2);
		System.out.println(gt1.get(1));
		} }
			C	1	Saha	Rahul	Compile time error	Runtime error

183	5	What is the output of the following code?
		
		import java.util.ArrayList;
		import java.util.List;
		
		public class Test {
		public static void main(String[] args) {
		List<String> al = new ArrayList<String>();
		al.add("A");
		al.add("B");
		al.add("A");
		al.add(null);
		System.out.println(al); } }
			D	1	Compile time error	[A, B, null]	[A, B]	[A, B, A, null]

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

184	5	What is the output of the following code?
		import java.util.ArrayList;
		import java.util.List;
		
		public class Test {
		public static void main(String[] args) {
		
		List<String> al1 = new ArrayList<String>();
		al1.add("A");
		al1.add("B");
		List<String> al2 = new ArrayList<String>();
		al2.add("P");
		al2.add("Q");
		al1.addAll(al2);
		System.out.println(al1); } }
			C	1	[A, B]	[P, Q]	[A, B, P, Q]	[P, Q, A, B]

185	5	What is the output of the code ?
		import java.util.ArrayList;
		import java.util.List;
		
		public class Test {
		public static void main(String[] args) {
		
		List<String> al = new ArrayList<String>();
		al.add("A");
		al.add("B");
		al.add("C");
		al.add(1, "Z");
		al.add(4, "V");
		System.out.println(al); } }
			A	1	[A, Z, B, C, V]	[A, B, C, Z, V]	[Z, A, B, V, C]	None of these

186	5	What is the output?
		import java.util.ArrayList;
		import java.util.List;
		
		public class Test {
		public static void main(String[] args) {
		List<String> al = new ArrayList<String>();
		al.add("A");
		al.add(4, "V");
		System.out.println(al);
		} }
			D	1	[A, V]	[A, null, null, null, V]	[A, 0, 0, 0, V]	IndexOutOfBoundsExcetption

187	5	Which of these method of ArrayList class is used to obtain present size of an object?	A	1	size()	contains()	capacity()	length()
188	5	Arraylist, Linkedlist and Vector are _____	A	1	classes	interfaces	packages	methods

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

189	5	What is the output of the Java code snippet?
		import java.util.*;
		class New {
		public static void main(String[] args) {
		ArrayList a1 = new ArrayList(50);
		a1.addAll(Arrays.asList("YES", "NO"));
		
		ArrayList<Integer> a2 = new ArrayList<Integer>(a1);
		a2.add(10);
		System.out.print(a2);
		}
		}
			B	1	[YES, NO]	[YES, NO, 10]	[10, YES, NO]	[YES]

190	5	In Java, the underlying data structure for the LinkedList class is?	B	1	Singly linked list	Doubly linked list	Circular linked list	None of these
191	5	LinkedList class implements?	C	1	List interface	Deque interface	both List & Deque interface	Vector & Stack

192	5	What is the output?
		import java.util.LinkedList;
		import java.util.List;
		public class Test {
		public static void main(String[] args) {
		List<Integer> ll = new LinkedList<Integer>();
		ll.add(100);
		ll.add(220);
		ll.add(50);
		ll.add(null);
		ll.add(100);
		ll.add(50);
		ll.add(100);
		ll.add(null);
		System.out.println(ll); } }
			C	1	[100, 220, 50, 100, 50, 100]	[100, 220, 50, null]	[100, 220, 50, null, 100, 50, 100, null]	None of these

193	5	What is the output?
		import java.util.LinkedList;
		
		public class Test {
		public static void main(String[] args) {
		LinkedList<Integer> ll = new LinkedList<Integer>();
		ll.add(100);
		ll.add(220);
		ll.add(50);
		ll.add(100);
		ll.add(50);
		ll.addFirst(999);
		ll.addLast(888);
		System.out.println(ll); } }
			C	1	[100, 220, 50, 100, 50]	[999, 100, 220, 50, 888]	[999, 100, 220, 50, 100, 50, 888]	[999, 100, 220, 50, 100, 50, 888,null]

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

194	5	What is the output of following code snippet?
		LinkedList<String> list = new LinkedList<String>();
		list.add("A");
		list.add("B");
		list.add("C");
		list.addFirst("D");
		list.addLast("E");
		list.add(2, "F");
		System.out.println(list);
			C	1	[D, A, B, F, C, E]	[D, A, B, C, F, E]	[D, A, F, B, C, E]	[D, A, B, C, E, F]

195	5	What is the output of following code snippet?
		
		LinkedList<Integer> list1 = new LinkedList<Integer>();
		list1.add(1);
		list1.add(2);
		LinkedList<Integer> list2 = new LinkedList<Integer>();
		list2.add(3);
		list2.add(4);
		list1.addAll(1, list2);
		System.out.println(list1.get(3));
			C	1	3	1	2	4

196	5	Which of the following methods is used to get the last element of a LinkedList in Java?	A	1	getLast()	getLastNode()	getLastElement()	getFirst()
197	5	Collection is _____ and Collections is _____.	A	1	Interface and class	class and Interface	In built framework & interface	class and class
198	5	What does the frequency() method of the Collections class do?	A	1	Returns the number of occurrences of an element in a List	Returns the number of occurrences of an element in a Set	Returns the number of occurrences of an element in a Map	None of the above

199	5	What is the output of the following code snippet?
		List<Integer> list = new ArrayList<Integer>();
		list.add(5);
		list.add(3);
		list.add(8);
		list.add(1);
		Collections.sort(list);
		System.out.println(Collections.max(list));
			C	1	5	1	8	3

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

200	5	What will be the output of the following code?
		import java.util.*;
		class xyz
		{
		public static void main(String[] args)
		{
		ArrayList a = new ArrayList(Arrays.asList(5 , 0 , 10 , 7 , 15 , 0 , 5 , 7));
		Iterator i = a.iterator();
		while(i.hasNext())
		{
		if((int) i.next() % 5 == 0)
		System.out.print(i.next());
		}
		} }
			C	1	55	5	0707	50101505

201	5	Which of the following is a valid way to reverse the order of the elements of a List using the Collections class?	C	1	list.reverse()	reversed(list)	Collections.reverse(list)	collections.reverse(list)
202	5	Which of the following is a valid way to find the smallest element in a List using the Collections class?	A	1	Collections.min(list)	list.min()	Collections.minimum(list)	list.minimum()
203	5	Which of the following is a valid way to find the frequency of an element in a List using the Collections class?	C	1	Collections.count(list, element)	list.count(element)	Collections.frequency(list, element)	list.frequency(element)
204	5	Which of these methods is used to add elements in vector at specific location?	A	1	add()	set()	AddElement()	addElement()
205	5	What is the correct syntax to declare a vector in Java?	A	1	Vector vec = new Vector();	Vector vec[];	Vector vec = {1, 2, 3};	None of the above.
206	5	Which method is used to add an element to the end of a vector?	A	1	add()	insert()	push()	append()
207	5	What is the default capacity of a vector in Java?	B	1	50	10	100	20
208	5	What is the correct syntax to create a vector of Strings in Java?	A	1	Vector vec = new Vector<String>();	vector<String> vec = new Vector();	vector vec = new Vector();	None of the above.
209	5	Which of the following methods is used to insert an element at a specific index in a vector?	B	1	add()	set()	addAll()	setAll()
210	5	Which is not correct way to get the last element of a vector in Java?	A	1	vec.get(0)	vec.get(vec.size() - 1)	vec.lastElement()	vec.elementAt(vec.size() - 1)

211	5	What is the output of the following code snippet?
		
		Vector<Integer> vec = new Vector<Integer>();
		vec.add(1);
		vec.add(2);
		vec.add(3);
		System.out.println(vec.get(1));
			B	1	1	2	3	4

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

212	5	What is the output of the following code ?
		
		import java.util.*;
		class vector
		{
		public static void main(String args[])
		{
		Vector obj = new Vector();
		obj.addElement(new Integer(3));
		obj.addElement(new Integer(2));
		obj.addElement(new Integer(6));
		obj.insertElementAt(new Integer(8), 2);
		System.out.println(obj);
		} }
			D	1	[3, 2, 6]	[3, 2, 6, 6]	[3, 2, 6, 8]	[3, 2, 8, 6]

213	5	What is the output of the following code snippet?
		Vector<Integer> vec = new Vector<Integer>();
		vec.add(1);
		vec.add(2);
		vec.add(3);
		vec.add(2);
		System.out.println(vec.indexOf(2));
			A	1	1	2	3	-1

214	5	What is the output of the following code snippet?
		Vector<String> vec = new Vector<String>();
		vec.add("Java");
		vec.add("Python");
		vec.add("C++");
		System.out.println(vec.get(2));
			C	1	Java	Python	C++	null

215	5	What is the output of the following code snippet?
		Vector<Integer> vec = new Vector<Integer>();
		vec.add(1);
		vec.add(2);
		vec.add(3);
		vec.remove(1);
		System.out.println(vec.get(1));
			C	1	1	2	3	null

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

216	5	What is the output of the following code snippet?
		Vector<Integer> vec = new Vector<Integer>();
		vec.add(1);
		vec.add(2);
		vec.add(3);
		vec.add(1);
		vec.add(2);
		vec.add(1);
		vec.add(2);
		vec.add(1);
		vec.add(2);
		vec.add(1);
		vec.add(2);
		vec.add(1);
		vec.add(2); System.out.println(vec.capacity());
			B	1	10	20	14	13

217	5	What will be the output of the following code?
		import java.util.*;
		class xyz {
		public static void main(String[] args) {
		Vector s = new Vector ();
		for(int i =0 ; i < 39 ; i++)
		{
		s.add(i);
		}
		s.add(2);
		s.add(null);
		System.out.print(s.get(39));
		System.out.println(s.capacity());
		}
		}
			C	1	240	3920	280	null40

218	5	What is the output of the Java code snippet?
		import java.util.*;
		class Test {
		public static void main(String[] args) {
		Vector v = new Vector<>(Arrays.asList(2, "LJ", 'h', 2, null));
		v.add("Easy");
		v.add(1, 0);
		v.remove('h');
		System.out.print(v.capacity() + v.size());
		System.out.print(v);
		}
		}
			D	1	17[2, 0, 2, null, Easy]	17[2, 0, LJ, 2, null, Easy]	Compile Error	Runtime Exception

219	5	Stack class is designed for ?	A	1	FILO Order	FIFO ORDER	LILO ORDER	None of the above

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

220	5	What will be the output of the following code?
		import java.util.Stack;
		
		public class Test {
		public static void main(String[] args) {
		
		Stack<String> st = new Stack<String>();
		st.push("Java");
		st.push("Kotlin");
		System.out.println(st.pop() + " " + st.pop() + " " + st.pop());
		}
		}
			D	1	Kotlin Java	Kotlin Java null	Compile time error	Exception at run time

221	5	What is the output of the following?
		import java.util.Stack;
		public class Test {
		public static void main(String[] args) {
		Stack<Integer> st = new Stack<Integer>();
		st.add(100);
		st.push(50);
		st.add(200);
		st.pop();
		st.add(220);
		st.push(50);
		System.out.println(st);
		} }
			C	1	[50]	[100, 50, 200, 220, 50]	[100, 50, 220, 50]	[220, 50]

222	5	What is the output of the following code?
		import java.util.Stack;
		
		public class Test {
		public static void main(String[] args) {
		Stack<String> st = new Stack<String>();
		System.out.print(st.isEmpty() + " ");
		st.push("Java");
		st.push("Kotlin");
		System.out.println(st.isEmpty());
		} }
			A	1	true false	false true	true true	false false

223	5	What is the output?
		import java.util.Stack;
		public class Test {
		public static void main(String[] args) {
		Stack<String> st = new Stack<String>();
		st.push("Java");
		st.push("Kotlin");
		st.push("Python");
		System.out.println(st.search("Java")
		+ " " + st.search("SQL"));
		} }
			B	1	1 0	3 -1	0 -1	3 0

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

224	5	import java.util.*;
		class stack
		{
		public static void main(String args[])
		{
		Stack obj = new Stack();
		obj.push(new Integer(3));
		obj.push(new Integer(2));
		obj.pop();
		obj.push(new Integer(5));
		System.out.println(obj);
		} }
			A	1	[3, 5]	[3, 5, 2]	[3, 5, 2, null]	[3, 2, 5]

225	5	What is the output of the following code?
		import java.util.Stack;
		class TestDemo2{
		public static void main(String[] args) {
		Stack s1 = new Stack();
		System.out.print(s1.size()+" ");
		s1.push("Hello");
		s1.push(20.23);
		s1.push("JJS");
		s1.peek();
		s1.peek();
		s1.peek();
		s1.pop();
		System.out.println(s1.size());
		} }
			A	1	0 2	0 1	compile time error	run time error

226	5	What will be the output of the following code?
		import java.util.*;
		
		class xyz {
		public static void main(String[] args) {
		Stack<String> st = new Stack<String>();
		st.push("Java");
		st.push("Kotlin");
		st.push("Python");
		System.out.println(st.search("java") + " " + st.search("SQL"));
		}
		}
			D	1	1 0	3 1	1 3	-1 -1

227	5	Write a program in which create two ArrayList. Add odd numbers between 1 to 10 in one arraylist and even numbers between 1 to 10 in another arraylist. Now merge these two ArrayList and sort them in descending order. At last print this sorted ArrayList.		2				
228	5	Write a Java program to create a new array list, add some colors (string) and print out the collection		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
229	5	Write a Java program that creates an ArrayList<Student> to store a name and marks of 3 students. Then sort the ArrayList by student marks in descending order. Then print arraylist using iterator. Add these details of 3 Students in arraylist in given sequence: Devarsh (18.5 marks), Viraj (20.5 marks) and Nidhi (17.5 marks) Desired Output: Name=Viraj, Marks=20.5 Name=Devarsh, Marks=18.5 Name=Nidhi, Marks=17.5		3				
230	5	Write a Java program to remove duplicates from an ArrayList as per given input and output. Take main class 'Exam'. Create ArrayList: [1, 2, 3, 4, 4, 5, 6, 7, 8, 9, 6, 9, 1, 1, 10] in main method then you need to implement another method named 'removeDuplicate' that takes an ArrayList as an argument , removes all duplicates and returns an ArrayList without duplicates. At end print that Arraylist from main method. (No need to use Scannner) Output of your code should be:- [2, 3, 5, 7, 8, 10]		3				
231	5	Remove Duplicates from an ArrayList Write a Java program to remove duplicates from an ArrayList. You need to implement a function that takes an ArrayList as input and returns an ArrayList with duplicates removed.		4				
232	5	Merge two sorted ArrayLists Write a Java program to merge two sorted ArrayLists into a single sorted ArrayList. You need to implement a function that takes two sorted ArrayLists as input and returns a single sorted ArrayList containing all the elements.		4				
233	5	Write a Java program to copy one array list into another. & sort the second arraylist & search an element in a array list		5				
234	5	Problem Statement: Employee Management System Implement an employee management system in Java using ArrayList and a class. The system should have the following functionalities: Add an employee to the system Remove an employee from the system Update an employee's details Display all employees in the system To implement this system, you can create an Employee class with attributes such as name, age, salary, etc. Then, you can create an ArrayList to store all the employees. You can then implement functions to add, remove, update and display employees.		5				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
235	5	Write a Java program that creates an ArrayList to store a list of students and their grades. The program should provide the following functionality: Add a student: The program should prompt the user for a student name and their grade, and add the student to the ArrayList. Remove a student: The program should prompt the user for a student name and remove the corresponding entry from the ArrayList. Search for a student: The program should prompt the user for a student name and output their corresponding grade. If the student name is not found in the ArrayList, the program should output an error message. Sort students by name: The program should sort the ArrayList by student name in ascending order. Sort students by grade: The program should sort the ArrayList by student grade in descending order. To accomplish these tasks, you can use the following Collections class methods: addAll(collection, elements): Adds all elements from one collection to another. sort(list): Sorts the elements in a list in natural order. sort(list, comparator): Sorts the elements in a list using a custom comparator. reverse(list): Reverses the order of the elements in a list. You can also use the following ArrayList methods: add(element): Adds an element to the end of the ArrayList. remove(element): Removes the first occurrence of an element from the ArrayList. get(index): Returns the element at the specified index in the ArrayList. size(): Returns the number of elements in the ArrayList.		6				
236	5	Problem Statement: Library Management System You are required to design a library management system in Java using ArrayList. The system should have the following classes: Book - This class should contain the following attributes: bookId, bookName, author, and quantity. It should also have a method to display book details. User - This class should contain the following attributes: userId, userName, and booksIssued. It should also have methods to display user details and to issue/return a book. Library - This class should contain an ArrayList to store all the books and an ArrayList to store all the users. It should have methods to add/remove books and users, to display all books/users, to issue/return a book, and to display all books issued by a particular user.		7				
237	5	Write a Java program to insert the specified element at the specified position in the linked list.		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
238	5	Write a Java program to get the first and last occurrence of the specified elements in a linked list.		3				
239	5	Write a Java program to clone an linked list to another linked list.		3				
240	5	Write a Java program to append the specified element to the end of a linked list & Write a Java program to iterate through all elements in a linked list.		4				
241	5	Problem Statement: Implement a Phone Book Write a Java program to implement a phone book using an ArrayList and a LinkedList. The phone book should allow users to add, delete, and search for contacts by name or phone number. To solve this problem, you can create a Contact class that stores the name and phone number of each contact. You can then use an ArrayList to store the contacts and a LinkedList to store the indices of the contacts sorted by name.		7				
242	5	I have an Induction machine manufacturing company. My Manager wants to analyze some data of the induction machines of our manufacturing unit. What I have: List of machine's price. Note: each price is in Double. Write a java program that contains list of various machine's price. & do the following tasks. Task 1 - Create a list of 5 machine's price. [10000.0,55000.0,80000.0,25000.0,55000.0] Task 2 - Find the maximum price from the list. Task 3 - sort the list.		5				
243	5	Write a program that contains Stack of 3 items. First insert 3 items in stack. Then do pop operation 2 times. and print the stack.		4				
244	5	Problem Statement: Balanced Parentheses Given a string containing only parentheses, determine if it is balanced. A string is considered balanced if every opening parenthesis has a corresponding closing parenthesis, and they are properly nested. You need to implement a function that takes a string as input and returns true if the string is balanced and false otherwise. HINT: To solve this problem, you can use a stack to keep track of the opening parentheses and pop them off the stack when a closing parenthesis is encountered. If at any point the stack is empty when a closing parenthesis is encountered or if the top of the stack doesn't match the current closing parenthesis, then the string is not balanced.		5				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
245	5	Write a Java program to sort a stack of Student objects by their GPA. You need to implement a function that takes a stack of Student objects as input and returns a new stack with the Students sorted in descending order of their GPA. To solve this problem, you can create a Student class with attributes such as name, id, and GPA. You can then create a stack of Student objects and implement a function to sort the stack by their GPA.		5				
TOPIC NAME:- 9 marks (PROGRAMS)								
246	4,5	Create a Java program for a Quiz game consisting of 5 Multiple Choice Questions (MCQs) with 3 participants. Each participant will be represented by the "Participant" class, which will have attributes such as "name," "age," and "contactNumber". To ensure the correctness of the contact number, implement a validation check. If a participant enters a contact number that is not 10 digits long or does not start with '9', display a message stating "Wrong number" and prompt the participant to enter the correct contact number. This validation check should continue until the participant enters a valid 10-digit contact number starting with '9'. Create objects of the "Participant" class and store them in an ArrayList. This ArrayList will hold all the participant objects participating in the quiz. Use a Stack data structure to store the correct answers to all 5 MCQs. Each answer should be represented as a character in the stack. This stack will serve as the reference for evaluating the participants' answers. The correct answer sequence is as follows: A, B , A , D , C. Create a thread for each participant using Java's threading mechanism. Each thread will represent an individual participant . The participant with the lowest age should be given the chance to answer all 5 questions first using scanner class. At a time only one participant can give answer of Questions. During the quiz, the participant's answers should be stored in an ArrayList. This ArrayList will hold the participant's submitted answers. After all participants have submitted their answers, calculate and display each participant's score. The score should be calculated by comparing		9				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
		each participant's submitted answers (ArrayList) with the correct answers (Stack). Award 1 point for each correct answer and calculate the total score for each participant out of 5. Finally, declare the winner based on the participant with the highest score. In case of tie, younger participant will be declared as winner. Below is the list of 5 Questions which is asked in Quiz. Q.1. National bird of INDIA..(A) Peacock (B) Sparrow (C) Duck (D) Owl Q.2 Independence year of INDIA..(A) 1955 (B) 1947 (C) 1999 (D) 1929 Q.3 Gandhi Jayanti is on.....(A) 2nd Oct (B) 5th Oct (C) 9th Oct (D) 7th Oct Q.4 Count of states in INDIA..(A) 17 (B) 21 (C) 25 (D) 28 Q.5 how many continents are there in the world..(A) 5 (B) 6 (C) 7 (D) 8 Answers: A, B , A , D , C						
		Filename: ConsProd.java You are tasked with implementing a multi-threaded program in Java that simulates the Consumer Producer problem. In this problem, you have a shared buffer between a consumer and a producer. The producer generates data and puts it into the buffer, while the consumer retrieves data from the buffer. Shared Buffer: Implement a SharedBuffer class. This class represents the shared resource between the producer and consumer threads. It should support operations for putting data into the buffer (one at a time by producer) and retrieving data from the buffer (one at a time by consumer). It has a value(int) field and isReady(boolean) field indicating whether the value is ready to be consumed or not. The produce() method sets the value and notifies the consumer, while the consume() method consumes the value and notifies the producer.						

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Computer Programming using Java-II PRACTICE BOOK (SEM-II-EVEN 2025-CE RELATED BRANCHES)

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
247	4,5	Producer: Create a Producer thread class that generates data items and puts them into the buffer. It repeatedly produces (total 5) values and sleeps for 1 second between each production. The producer should wait if the buffer is full. Consumer: Create a Consumer thread class that retrieves data items from the buffer. It repeatedly consumes the produced value (total 5) and sleeps for 1 second between each consumption.The consumer should wait if the buffer is empty. Main Class (ConsProd): This is the entry point of the program. Create an object of the SharedBuffer class and two threads - one for the producer and one for the consumer. Then start both threads to achieve goal. Sample output: (assume above mentioned delay between each production and each consumption) Produced: 1 Consumed: 1 Produced: 2 Consumed: 2 Produced: 3 Consumed: 3 Produced: 4 Consumed: 4 Produced: 5 Consumed: 5 Properly do use of synchronization in methods of the SharedBuffer class to ensure thread safety, and wait() and notify() methods for coordinating between the producer and consumer threads.		9				